MIL-C-50534A (MU)
15 January 1971
SUPERSEDES
MIL-C-50534 (MU)
25 May 1970

MILITARY SPECIFICATION

CASE, CARTRIDGE, XM205E2, 152MM, PARTS FOR

1. SCOPE

1.1 This Specification covers the component parts of the cartridge case designated as Case, Cartridge, 152MM, XM205E2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this Specification to the extent specified.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-131	-	Barrier Material, Water-Vaporproof, Flexible
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-I-45607	-	Inspection Equipment; Supply and Maintenance, For Ordnance
MIL-C-50269	-	Cellulose, Woodpulp (Sulfate) for nonmetallic Cartridge Cases, Kraft
MIL-F-50533	-	Fiber, Acrylic, Fibrillated
MIL-P-81705	-	Film, Transparent, Flexible, Electrostatic-Free, Heavy Duty

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-650	-	Explosive: Sampling, Inspection & Testing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-	Lot Numbering of Ammunition

FSC1320

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FEDERAL

FED-STD-406 - Plastics; Method of Testing

DRAWINGS

ARMY

8799112	-	Marking Diagram and Sealing of Ammunition Fiberboard Containers
8889489	-	Box Fiber, for Body, for Case, Cartridge, XM157E1 or XM205E2
9242503	-	Box, Fiber, for Base, for Cartridge Case, XM157E1 or XM205E2
9252187	-	Base
9252186	-	Body
9247366	-	Composition
9252619	-	Case, Cartridge Assembly, 152MM, XM205E2

PUBLICATIONS

ARMY

ET-9252619 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS, Title 49: Transportation; Parts 49 CFR 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington D.C., 20402. Orders should cite "49 CFR 0-190 (latest revision).")

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM-D-638 - Tensile Properties of Plastics

(Copies may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia Pennsylvania, 19103).

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3. REQUIREMENTS

3.1 Materials.-The constituent materials shall comply with the requirements of all applicable drawings and specifications (see 6.4).

3.2 Parts.-The cartridge case parts shall comply with all the requirements specified in Drawing (dwg.) 9252187 and 9252186 and with all the requirements specified in the applicable specifications.

3.3 Stability.-The nitrocellulose in the cartridge case body and base shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested according to 4.3.3.1.

3.4 Physical properties.-The cartridge case parts shall comply with the requirements of Table I when tested in accordance with 4.3.3.2 and 4.3.3.3.

TABLE I
Physical Properties

Item	Weight	Tolerance	Tensile Strength
Body	295 gms	± 12 gms	2500 psi Minimum (min)
Base	74 gms	± 4 gms	2200 psi (min)

3.5 Workmanship.-All parts shall be fabricated in a thorough, workmanlike manner, and shall be free of molding flaws, cracks, warpage outside the material envelope, crevassing, soft spots, frayed edges, foreign material greater than 1/4 inch in any dimension and delaminations.

3.6 Inspection fixture gage.-The cartridge case body and base shall enter freely and position properly on a fixture gage. Testing shall be in accordance with 4.4.3.

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3.7 Proving ground.-The cartridge case, when assembled as part of a 152MM cartridge, shall function satisfactorily without smoldering residue, and without more than one occurrence of excessive residue (see 6.6) when tested in accordance with 4.4.4.

3.8 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as apecified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified here, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-When an inspection lot of cartridge cases (bodies and bases together) is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the inspection lot of cartridge cases complies with all of the quality assurance provisions specified within this specification.
b. A statement that the components included in the inspection lot of cases (are) (are not) hybrid (see 6.10).
c. Number of components in the inspection lot.
d. Number of components in the inspection lot that were inspected.
e. Results obtained, by defect code, for all inspections performed.
f. Drawing, specification number, and date, together with identification and dates of changes.
g. Certificates of conformance.
h. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Examination and ballistic tests

4.2.1.1 Submission.-The contractor shall submit a first article sample, as designated by the contracting officer, for evaluation in accordance with the provisions of 4.2.1.2. The first article sample shall consist of 150 complete sets of parts (i.e., 150 cartridge case bodies and 150 cartridge case bases) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures, molds, dies (see 6.7) and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government).

4.2.1.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified in 4.4.

4.2.2 Rejection.-If any component part fails to comply with the requirements for the inspections performed in accordance with 4.2.1.2, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection the Government reserves

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the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until the initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the contracting officer.

4.2.3 New equipment approval.-Prior to the use of a new mold or punch die in regular production of the case body or base, the contractor shall submit to the Government five (5) parts produced from the new mold or punch die for inspection. If any part fails to comply with the applicable drawings or specification requirements, when tested as specified in 4.2.1.2, the mold or die shall not be approved for use in regular production.

4.3 Inspection provisions

4.3.1 Lot formation

4.3.1.1 Case bodies and case bases.-The term "lot", as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous (see 6.9) collection of bodies or bases from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same specification revisions. Changes to the process, specification, or drawing not affecting safety, performance interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. A maximum of ten percent of the total number of items deliverable under the contract may be submitted as hybrid lots (see 6.10). Inspection lots shall comply with Standard MIL-STD-105. Inspection lots shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with Standard MIL-STD-1168. Each lot of case bodies or case bases shall contain parts produced from:

a. Nitrocellulose from one manufacturer and from not more than one production lot.

b. Diphenylamine from one manufacturer and from not more than one production lot.

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c. Acrylic fibers from one manufacturer and from not more than one production lot.
d. Resin from one manufacturer.
e. Kraft from one manufacturer and from not more than one production lot.

4.3.1.2 Case, Cartridge.-A cartridge case lot shall consist of case bodies and bases. Each cartridge case lot shall contain:

a. No more than one lot of case bodies
b. No more than one lot of case bases (see 6.10)

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be in accordance with MIL-STD-105 except that continuous sample plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.3. Conditioning shall be at 23 ± 2°C and 50 ± 5% R.H. for a minimum of 24 hours.

4.3.2.1 Body, for Case, Cartridge, XM205E2 (see dwg. 9252186).

Category	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Length from shoulder to aft end	See 4.4.3	01001
102.	O.D. of aft end (max)	See 4.4.3	01002
103.	Dome Height	See 4.4.3	01003
104.	Contour of Dome (max)	See 4.4.3	01004
105.	True position of mouth	See 4.4.3	01005
106.	Runout of aft end	See 4.4.3	01006
107.	Perpendicularity of shoulder	See 4.4.3	01007
108.	O.D. of aft end (min)	Gage	01008
109.	Contour of dome (min)	Gage	01009

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Minor:	AQL 4.00 percent		
201.	Small diameter of mouth (min)....	See 4.4.3	01010
202.	Large O.D. at shoulder.....	Gage	01011
203.	Wall thickness.....	Gage	01012
204.	Depth of scallops (min).....	Gage	01018
205.	Number of scallops.....	Visual	01019
206.	Presence of tabbing.....	Visual	01020
207.	Workmanship (see 3.5).....	Visual	01013
208.	Presence of chamfer on mouth diameter.....	Visual	01021

4.3.2.2 Base for Case, Cartridge, XM205E2 (see dwg. 9252187).

Category	Defect	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	I.D. (min).....	See 4.4.3	02001
102.	Depth of primer cavity.....	See 4.4.3	02002
103.	True position of primer cavity...	See 4.4.3	02003
104.	True position of large I.D. to large O.D.....	See 4.4.3	02004
105.	Large O.D. (max).....	See 4.4.3	02006
106.	Large O.D. (min).....	See 4.4.3	02019
107.	Diameter of primer cavity.....	Gage	02005
108.	Radius of forward outside corner (max).....	Gage	02020
Minor:	AQL 4.0 percent		
201.	Height of lip.....	See 4.4.3	02007
202.	Length from inside flat surface to apex.....	See 4.4.3	02008
203.	Thickness of lip (min).....	Gage	02009
204.	Diameter of boss.....	Gage	02010
205.	Thickness of base end on flat surface.....	Gage	02011
206.	Diameter of flash hole.....	Gage	02012
207.	Absence of inspection hole.....	Visual	02013
208.	Workmanship (see 3.5).....	Visual	02014

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4.3.2.3 Packaged case body or case base, prior to sealing of packing box.

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 100 percent inspection		
101.	Protective packaging missing or torn	Visual	03001
102.	Heat seal incomplete on protective packaging	Visual	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship in assembly of protective packaging	Visual	03003

4.3.3 Testing

4.3.3.1 Stability (see 3.3, Major defect (Body Code No. 01014; Base Code No. 02015).-A stability heat test at 134. 5°C shall be performed in accordance with MIL-STD-286, Method 404.1.2. Random samples of 10 case bodies and 10 case bases shall be selected from the first article sample and the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to 5 bodies and 5 bases from each lot. However, if a failure occurs, sampling shall revert to 10 bodies and 10 bases per lot until 3 consecutive lots have been accepted.

4.3.3.2 Tensile strength (see 3.4), Major defect (Body Code No. 01015; Base Code No. 02016.-Ten case bodies and ten case bases shall be randomly selected from the first article sample and from each of the first 3 production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When 3 consecutive lots have been accepted, sampling may be reduced to 5 bodies and 5 bases from each lot. If a failure occurs, sampling shall revert to 10 bodies and 10 bases per lot until 3 consecutive lots have been accepted. Two specimens shall be prepared from each body or base, taken from the center of the wall of the body, and from the flat surface of the base. Test specimens shall be taken 180° apart. The specimens shall have a one-inch gage length and width, but wall thickness shall fall within the tolerance specified on the applicable drawing. The test specimens shall be preconditioned for a minimum of 4 hours at 23 plus or minus (+) 2° Centigrade (C) at a relative humidity of 50 ± 5 percent (%). The tensile breaking strength of the cartridge case body and base shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute.

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4.3.3.3 Weight (see 3 4), Major defect, (Body Code No. 01016: Base Code No. 02017.-Samples shall be selected in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235). AQL for this defect shall be 0.40%. All samples shall be weighed individually, using any Government approved and suitably calibrated scale capable of reading to the nearest 0.1g. Test samples shall be preconditioned for a minimum of 4 hours at 23 ± 2 degrees C and 50 ± 5% R.H. prior to weighing.

4.3.3.4 Composition (see dwg. 9247366), Major defect (Body Code No. 01017: Base Code No. 02018) (see 3.1).-Ten case bodies and ten case bases shall be randomly selected from the first article sample and from each of the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to five bodies and five bases from each lot. If a failure occurs, sampling shall revert to ten bodies and ten bases per lot until three consecutive lots have been accepted. Duplicate specimens shall be obtained from each sample for analysis in accordance with 4.4.1.

4.3.3.5 Density Informational.-Five bodies and five bases shall be randomly selected from the first article sample and from each of the first three production lots to provide the test specimens for this test. One specimen shall be taken from the flat surface of the base end and one specimen shall be taken from the lip of each sample case base. From each sample case body, one specimen shall be taken from the forward end of the sidewall and one specimen shall be taken 180 degrees away at the aft end of the sidewall. The density of each specimen shall be determined in accordance with 4.4.2 for information (see 6.11).

4.3.3.6 Inspection fixture gage test (see 3.6).-Case bodies and bases shall be randomly selected from each lot in accordance with Standard MIL-STD-105, and subjected to an inspection fixture gage test. Conditioning of all samples shall be at 23 ± 2°C and 50 ± 5% R.H. for 24 hours minimum prior to inspections required by 4.3.2.1 and 4.3.2.2.

4.3.3.7 Proving ground (see 3.7), Smoldering residue/Special defect, Code No. 04001; Excessive residue/Major defect, Code No. 04002.-One hundred twenty (120) cartridge cases (see 4.3.1.2) shall be randomly selected from each lot until three (3) lots have been accepted forty (40) case bodies and forty (40) case bases shall subsequently be randomly selected for proving ground testing. Sample cases shall be assembled to 152MM projectiles and ballistically tested as specified in 4.4.4. Failure to comply with the applicable requirement when tested at 70° Fahrenheit (F) shall be cause for lot rejection. Firings at -40°F and +125°F shall be for information only (see 6.8).

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4.3.4 Inspection equipment.-Equipment Tabulation number ET-9252619 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for performance of the examination or test by the Government.

4.3.4.2 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used for DOD activities for design, procurement, manufacture, testing, evaluation, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Composition

4.4.1.1 Preparation of specimen (see 4.3.3.4).-Each specimen shall be divided into 1/8-inch (approximate) cubes or equivalent. The specimens shall then be thoroughly mixed to form a composite. From the composite, approximately 1.0 g. shall be selected as the analysis sample and weighed to the nearest 0.2 milligram (mg), (Weight A) in an extraction thimble. The same thimble should be carried through the extractions in 4.4.1.3.1 and 4.4.1.3.3.

4.4.1.2 Total volatiles.-The samples from 4.4.1.1 shall be dried for 3 hours at 100 $\pm$ 2°C, cooled in a desicator (1/2 hr. min.), and weighed to the nearest 0.2 mg (Weight W). Calculate the percent total volatiles as follows:

$$\% \text{ Total Volatiles} = \frac{A - W}{A} \times 100$$

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4.4.1.3 Analytical procedure

4.4.1.3.1 Extraction.-Using a tared, extra-coarse, sintered glass thimble, extract each sample from 4.4.1.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286, Method 104.1). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.4.1.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. Dry the thimble plus residue for 1/2-hour minimum at 100 ± 2°C. Cool in a desiccator (1/2 hr. min). Determine the dry weight of this residue (Weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.4.1.3.3..

4.4.1.3.2 Diphenylamine (DPA).-To the dried extract from 4.4.1.3.1, add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with paragraph 5.5) of Standard MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in Standard MIL-STD-286, Method 201.4, paragraph 5.11.

4.4.1.3.3 Nitrocellulose.-To provide good heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.4.1.3.1 and extract with acetone using the method in 4.4.1.3.1. After extraction dry the residue remaining at 100 ± 2°C for one hour. Cool in a desiccator (1/2 hr. min) weigh to the nearest 0.2 mg (Weight D) and save for the acrylic determination in 4.4.1.3.4. Evaporate the acetone extract to dryness and add 25 ml of n-butyl acetate and 45 ml of glacial acetic acid. Replace the air with a stream of inert gas (CO_2 or N_2). Maintain the inert atmosphere throughout this test. Reflux the solution for five (5) minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1) and 10 ml concentrated hydrochloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the specimen is

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completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammonium thiocyanate solution (MIL-STD-286, Method 601.1). Titrate with a 0.2N titanous chloride solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate the percent NC as follows:

$$\% \text{ NC} = \frac{46.69N \quad (V-B) \quad X\ 100}{E\ W}$$

Where:

V = Volume titanous chloride used for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.4.1.3.4 Acrylic

Reagents

SULFURIC ACID - 93 - 98% H_2SO_4, N_2 - Free

MERCURIC OXIDE (HgO) - Reagent Grade, N2 Free

POTASSIUM SULFATE or ANHYDROUS SODIUM SULFATE - Reagent Grade, N_2 Free

THIOSULFATE SOLUTION - 160 g $Na_2S_2O_3 \cdot 5H_2O$ dissolved in one liter of H_2O

SODIUM HYDROXIDE - 400 g solid NaOH dissolved in H_2O and diluted to one (1) liter

METHYL RED INDICATOR - 1 g methyl red dissolved in 200 ml ethyl alcohol

METHYLENE BLUE INDICATOR - 1 g methylene blue dissolved in 200 ml ethyl alcohol

HYDROCHLORIC or SULFURIC ACID STANDARD SOLUTION - 0.1N

BORIC ACID SOLUTION - 4 g H_3BO_4 dissolved in 100 ml H_2O

HENGAR GRANULES

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Apparatus

DIGESTION-Use Kjeldahl flasks of hard, moderately thick, well-annealed glass with total capacity of 500 to 800 ml. Conduct digestion over a safety-approved heating device adjusted to bring 250 ml H_2O at 25°C to a rolling boil in 5 minutes.

DISTILLATION-Use a Kjeldahl flask of 500 to 800 ml capacity. fitted with a rubber stopper. The low end of an efficient scrubber bulb or trap is passed through the rubber stopper to prevent mechanical carryover of NaOH during distillation. Use rubber tubing to connect the upper end of the bulb to the condenser tube. Trap the condenser outlet to insure complete absorption of distilled NH_3 into boric acid in the receiver.

Analysis

Quantitatively transfer the residue from 4.4.1.3.3 to a Kjeldahl flask. Add 3-4 Hengar granules, 0.7g of HgO, 15.0g of K_2SO_4 or Na_2SO_4, and 25 ml H_2SO_4 to the flask. Place the flask in an inclined position, and heat gently until frothing ceases. Boil the contents of the flask briskly until the solution clears. Continue boiling for 1/2 hour minimum. Allow the solution to cool to room temperature. Add 200 ml H_2O. Cool the solution to below 25°C.

Add 25 ml thiosulfate solution and mix to precipitate the mercury. Tilt the flask and add a layer of 100 ml NaOH solution without agitation. Immediately connect the flask to the distilling bulb on the condenser. With the tip of the condenser immersed in the boric acid in the receiver, rotate the flask to mix the contents throughly. Heat the flask until all of the NH_3 has distilled, (approximately 250 ml). Titrate the boric acid solution with the standard acid using a mixed indicator containing 4 drops of methyl red, and 2 drops of methylene blue solution. Correct for the reagent blank. Calculate the percent acrylic as follows:

$$\% \text{ Acrylic} = \frac{6.05 \quad (V-B) \quad N}{W} \quad X\ 100$$

Where:

V = Volume of standard acid for sample, ml
B = Volume of standard acid for blank, ml
N = Normality of standard acid
W = Dried sample weight, g

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4.4.1.3.5 Kraft, resin and additives.-Calculate the percent kraft, resin and additives as follows:

$$\%\text{Kraft} + \%\text{resin} + \%\text{additives} = 100 - (\%\text{NC} + \%\text{Acrylic} + \%\text{DPA})$$

4.4.1.3.6 Report the total methylene chloride extractables $\frac{(W - C)}{W} X\ 100$ and the total acetone extractables $\frac{(C - D)}{W} X\ 100$

4.4.2 Density.-The density of each specimen shall be determined by the mercury displacement method in accordance with MIL-STD-650, Method 203.1.

4.4.3 Inspection fixture gage testing.-Examinations in 4.3.2 shall be conducted in accordance with instructions contained on the applicable Inspection Equipment Drawing (see 4.3.4). Observations shall be made for compliance with the requirement of 3.6.

4.4.4 Proving ground.-Testing shall be in accordance with the applicable Proving Ground Acceptance Test Procedure. Testing shall be at a Government Proving Ground designated by the contracting officer. The test samples shall be fired at the temperatures indicated in Table II, and accepted in accordance with Table III.

Table II

Temperature	First Article	First 3 Lots	Subsequent Lots
- 40°F	20	20	--
+ 70°F	80	80	40
+125°F	20	20	--

Table III

Sample Size	Acceptance Criteria for Smoldering Residue		Acceptance Criteria for Excessive Residue	
	Accept	Reject	Accept	Reject
80	0	1	1	2
40	0	1	1	2

* All test firings at - 40°F and +125°F shall be for information only.

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Observation shall be made for compliance with the applicable requirement, 3.7. Camera coverage of the breech block and chamber of the weapon shall be provided for all test firings.

4.4.4.1 Refiring.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the results shall be declared invalid, and a new test shall be performed.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level C.-Each case body shall be sealed in a 4 mil (minimum thickness) polyolefin bag. A maximum of twenty (20) case bases shall be packed together in a similar polyolefin bag. The material shall be black with surface resistivity not greater than 1000 ohms per centimeter when determined in accordance with Standard FED-STD-406, Method 4041 and 4052. The electrostatic bleed-off shall comply with the requirements of Specification MIL-P-81705, paragraph 4.9.9. The method of sealing shall be a Government-approved heat seal.

5.2 Packing

5.2.1 Level C.-Bagged case bodies and case bases shall be packed in accordance with dwg. 8889489 or 9242503, as applicable.

5.3 Marking.-Unless otherwise specified, markings shall be in accordance with dwg. 8799112, and dwg. 9252619.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each case inspection lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection Equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Equipment Lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock Numbers of existing equipment designs or codes, as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design or controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commerical equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This required that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance

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with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, Contractor design may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contract reproductions. Contractors shall submit two copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Additives shall have prior approval by the contracting officer. Mold release agents and processing aids are considered additives.

6.5 Submission of first article sample.-Instructions as to the location for the evaluation of first article samples shall be obtained from the contracting officer, upon request. All inquiries should be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NK1
Dover, New Jersey 07801

6.6 Excessive residue.-Residue is defined as excessive if it is significant enough to prevent chambering or hinder functioning of a subsequent round, or exceed 1 inch by 2 inches in size, as determined by the proving ground.

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6.7 First article sample.-The first article sample shall contain a representative quantity (a minimum of five (5) of bodies and bases fabricated from each mold, and punch die intended for use in regular production. Each body and base sample shall be appropriately labelled as to mold and punch die used during its fabrication.

6.8 Proving ground

6.8.1 Assembly of cartridge case.-Cartridge cases shall be assembled to 152MM XM411 or equivalent projectiles using normal loading procedures, except that no adhesive or lacquer shall be used to cement the case body to the case base, or in the assembly of the Primer.

6.8.2 Cleaning.-Prior to each firing, the breech and firing probe of the test weapon shall be thoroughly cleaned and dried.

6.8.3 Handling and storage.-Special care shall be taken in the handling and storage of sample cartridge cases to prevent contamination by grease, grime, dirt, or other foreign material.

6.9 Homogeneous lots.-A component lot shall be considered homogeneous if the lot contains source materials in compliance with 4.3.1.1.; i.e., contains no more than one (1) lot of nitrocellulose, one (1) lot of woodpulp, one (1) lot of acrylic, and one (1) lot of DPA. An inspection lot of cartridge cases shall be considered homogeneous if it contains only one (1) homogeneous body lot and one (1) homogeneous base lot. The body and base components are not required to be of the same component lot designation or of the same lot number as the case lot. However, only one (1) body and one (1)base lot is permissible in a case lot as defined in 4.3.1.2.

6.10 Hybrid lots.-A component lot shall be considered hybrid if the lot contains source materials which are not in compliance with 4.3.1.1; i.e, the lot contains more than one (1) lot of NC, Acrylic, woodpulp, or DPA. An inspection lot of cartridge cases shall be considered hybrid if: (1) the lot contains a body lot or base lot that is hybrid and/or (2) the lot contains more than one (1) body lot or base lot. In either condition of (1) or (2), the entire case lot shall be considered hybrid.

6.11 Submission of data.-One copy of all data generated from informational tests shall be forwarded to SMUPA-ND2 and SMUPA-DE3.

Custodian:
Army-MU

Preparing Activity:
Army-MU

FSC: 1320

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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